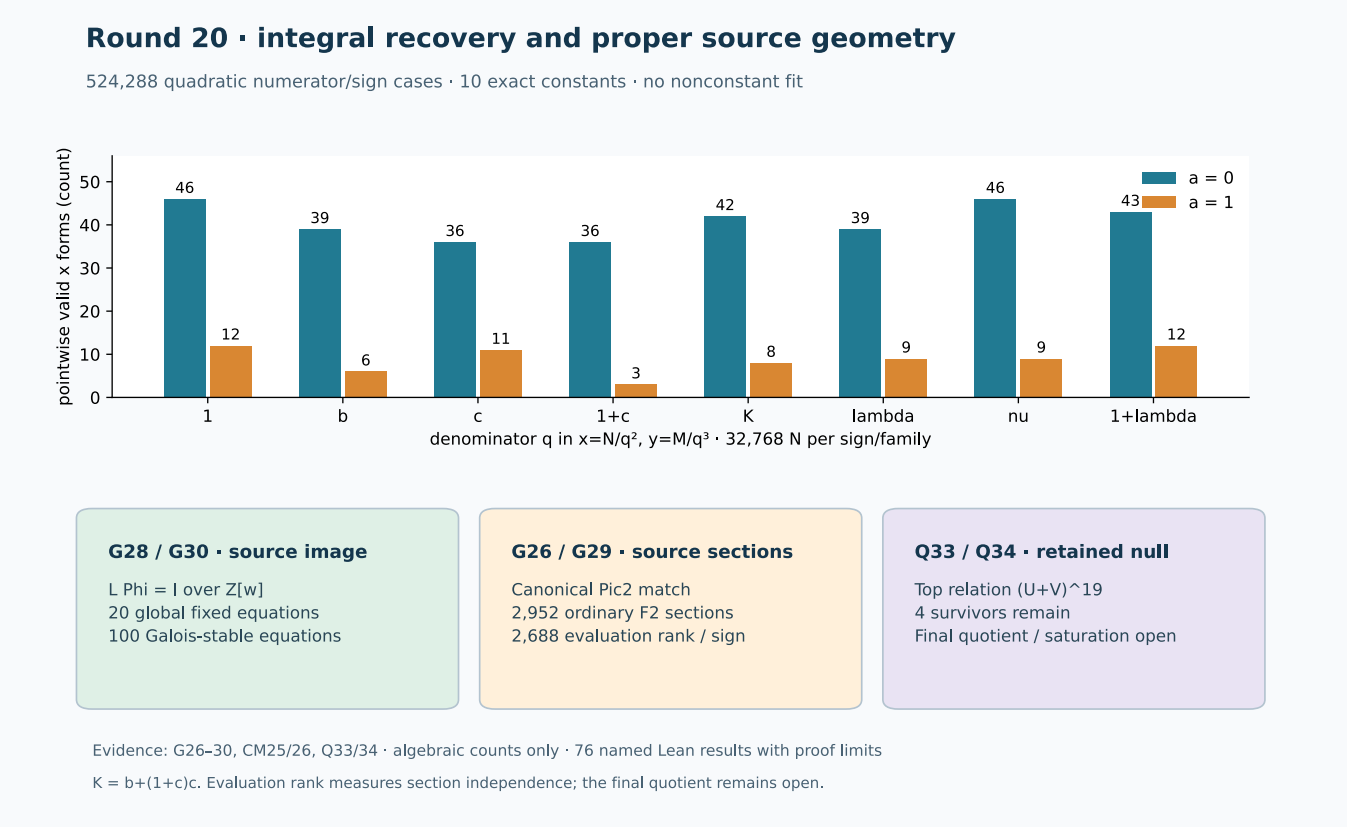
Round20: integral recovery and proper source geometry

24 October 2026 America/Los_Angeles; execution24 October UTC. Scientific parent11b2f23,operational parentef4f01;Conductor T-291. Frozen R01–19 artifacts are unchanged. This round continues the integral level77 map, canonical mixed theta sections and full quotient saturation. The user retains assessment of IC efficiency.

The new evidence is an explicit integral CM recovery map for all20 G25 row maps. It proves a proper embedding of the scalar105 source under the written group/descent bridges,and supplies20 global proper source-image equations,or100 in a Galois-stable orbit. Canonical symmetric Pic2 matching holds. A factorized complete elliptic tensor embedding is specified; 2952 sparse ordinary F2 sections are evaluated. The integral modular map, final quotient243-space and whole saturation remain open.



requirement	result and exact scope
canonical mixed class	G26:actual bias[1,1,1,1] matches both corrections;only the zero Pic0[2] vector is Frobenius fixed;written line-bundle/descent proof
proper source geometry	G28:L*Phi=I4 over Z[w],w²+w+2=0;literal4x20 recovery,max17 coefficient magnitude bits;eight actual tuple round trips
proper equations	G30:E=Phi*L,E²=E;20 global elliptic-group fixed equations;five conjugates,100 Galois-stable equations
full factor embeddings	G29:complete degree45,315,63,171 elliptic bases and degree57 companion;factorized tensor/Segre embedding of the source
mixed F2 sections	2952 ordinary sections;16 tuples per sign,168-bit evaluations;binary rank2688 each;48 scalar-kernel noninvariance witnesses
integral level77 map	CM25/26:524288 complementary quadratic numerator/sign cases,10 finite fits,all exact constants;no unexpanded spaces
quotient saturation	Q33:186-term norm per graph,top(U+V)^19;does not exclude Q32;Q34 keeps four complete uncapped models on123,124,126,127

The section rank is a lower bound on independence over F2,not an IC relation rank or a source-space dimension. These experiments measure algebraic counts and exact identities,not runtime gains. No external-CAS speed comparison or IC efficiency verdict is supplied. See THETA.md,CM.md and SATURATION.md for written arguments and the exact unresolved obligations.

Geometry,sections and retained failures

Ordinary factors are K_a:y^2+xy=x^3+a*x^2+1 over F2,a=0 or1. The geometric source is the n=5 trace-zero fourfold on the scalar105 refinement,with G15's unchanged supersingular companion. Exact point-field models and public synthetic controls are the frozen G17/G22 and CM17/22 inputs;no unknown-scalar recovery is run.

G27's actual prime3,5,7 matrices have rank8 and explicit left inverses for both signs,with960 actual point-column controls. G28 strengthens that finite evidence to an integral retraction. Starting from U0*M=105I in the 40x8 integer row lattice,three exact prime lift steps yield U*M=I. Every intermediate lift and all CM coefficients are retained. Lean proves the actual coefficient recovery and its universal evaluated-ring implication; applying it to elliptic group schemes is a written bridge.

For the proper source image,set E=Phi*L. Every fixed Q is Phi(LQ),and every Phi(P) is fixed. Thus the global equalizer is exactly the source on every test scheme. The resulting20 morphism equations include infinity without an affine saturation assumption. The companion map is the identity. These are group-morphism equations,not an expanded small homogeneous ideal or the final105/19 quotient equations.

Each full elliptic basis contains x,y and consecutive infinity-pole coordinates with ratio x/y. Product/Segre gives a proper polarized source embedding. The ordinary tensor coordinate count is exactly 830421600035681310437325756310786230468 75;with the companion it is 473340312020338346949275681097148151367187 5. This is a symbolic Cartesian system,not a claim that its restriction spans the source H0 of dimension 129606823895507929687 5. The canonical class match still lacks an explicit rational trivializing function.

The sparse 2,952-section system uses all five orbit coefficients for every nonconstant factor-basis function,plus1 and the product of four highest- pole orbit products. All190 row pairs are CM independent;under the written divisor bridge the top product has exact pole divisor D,removing the old forced ordinary fixed divisor. Codimension2 base loci remain open. All48 old scalar-kernel translations remain noninvariant;complete kernel actions and243 final quotient sections are missing.

Q33's initial export incorrectly advertised rejection although the native boundary residue was zero. That export was never built or used as a proof; its source and output are retained. The actual denominator polynomial is constant1,not degree1. The corrected relation gives only U=V on the reduced boundary,already satisfied by all four Q32 points. Q34 retains every prior constraint and finds the same four models in nine visits per graph. No complete simultaneous source arc is constructed.

CM25 searches all F2 quadratic N and all F32 quadratic y numerators for eight denominators on18 actual incidences. The ten finite fits are independently constant as free polynomials,N=x0*q^2 and M=y0*q^3. This is a complementary bounded ansatz,not the prescribed full Riemann–Roch space. Proper normalization,pole divisors and the integral factor/divisor evaluator remain missing. Other retained failures include the initial vacuous G27 nullspace proof,replaced by actual left inverses;native type errors;the wrong-directory Lean launch;and the root F4 cost-threshold failure.

Validation and next discriminating work

PROOF_MAP.md maps 76 new named Lean statements to their exact hypotheses. The complete project build passes 9,061 jobs;the transitive audit passes 821 theorems with zero failures and zero trusted native results. VALIDATION.md and ledger.json record the completed checks. The initial cache preflight is interrupted on NFS;the exact full build and audit stages pass using the existing pinned dependencies. Written Picard/divisor/group-scheme/descent arguments,native field semantics, search exhaustion and evaluation ranks are kept distinct from kernel checked results. Only standard foundations are allowed;no sorry,custom axioms or trusted native evaluation is accepted.

Root release tests pass 4,493 with 119 ignored on the full rerun;the first failure and isolated passing control are retained. Native tests,site and workflow/budget checks,standalone Lean control,identity checks,member hashes, source pins and exact replay are recorded in VALIDATION.md and runs/COMMANDS.md. Three source-bound universal identities live beside this report. REPLAY.sh reruns16 native commands and compares16 JSON outputs and8 generated Lean modules. Deterministic raw and validation archives require fresh extraction, every member hash and byte-identical repacking;see ARCHIVE_VALIDATION.txt.

Both archives pass fresh member hashes,native manifests,full byte replay and identical repacking:198 raw files and253 validation files,207 current source pins and190 historical R19 pins. Archive sizes are2989460 and 5623828 bytes,respectively,both below10 MiB. The completed receipts are in ARCHIVE_VALIDATION.txt.

The migration receipt,Diem owner and existing EKS inventory were reconciled before experiments. Diem is deadline-expired with50/52 durable records and no pending work;no duplicate generator or EKS job was launched. The separate source-side waiting-freeze followup was not relaunched. No prior completed R19 arithmetic was repeated merely for recovery.

The next geometry obligation is explicit rational canonical trivialization and complete scalar/final19 kernel actions on the weighted sections,with a verified invariant243-space and proper quotient equations. The next saturation test is a complete simultaneous source arc satisfying every inverse,Kummer and coefficient-Frobenius block,or an exact source-bound separator rejecting a survivor. The modular map needs normalization and actual coordinate pole divisors before a complete controlled-pole space can replace the bounded ansatz. The460/367 ordinary-support source fibers and images also remain unresolved.

Canonical study/frontier,problem,backlog and dashboard coverage are refreshed with these distinctions. Historical relation ranks,IC costs,rho ratios and performance plots are unchanged:there is no new verified relation matrix or performance measurement. The round's diagram and count plot are generated from the retained native records;report.pdf is inspected page by page. Primary sources and their limited written use are listed in SOURCES.md.

The source-bound identity IDs are IDC1h8695f783c1779eb3 for the integral lift step,IDC1h15ad1abfa890dd5b for the universal power19 factorization,and IDC1h5b1912d4dba97ee6 for coordinate clearing. The characteristic2 sparse expansion is proved separately in Lean;the odd-prime certificate does not certify that characteristic-specific statement.